

## Worksheet 2.1

**Student Name:** Ayush Yadav

**Branch:** MCA (AI & ML)

**Semester:** II

**Subject Name:** Machine Learning Lab

**UID:** 25IMC13004

**Section/Group:** MAM-1 (A)

**Date of Performance:** 23/02/26

**Subject Code:** 25CAP-672

### 1. Aim:

Write a program for Support Vector Machine using python.

### 2. Objective:

To implement a **Support Vector Machine (SVM)** algorithm using Python and scikit-learn library for classification of a dataset, and to evaluate its performance using accuracy and other performance metrics.

### 3. Coding:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
data = pd.read_csv("Social_Network_Ads.csv")
```

```
X = data[['Age', 'EstimatedSalary']].values
```

```
y = data['Purchased'].values
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
model = SVC(kernel='linear')
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```



```
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")
```

```
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
x_min, x_max = X_train[:, 0].min() - 1, X_train[:, 0].max() + 1
y_min, y_max = X_train[:, 1].min() - 1, X_train[:, 1].max() + 1
```

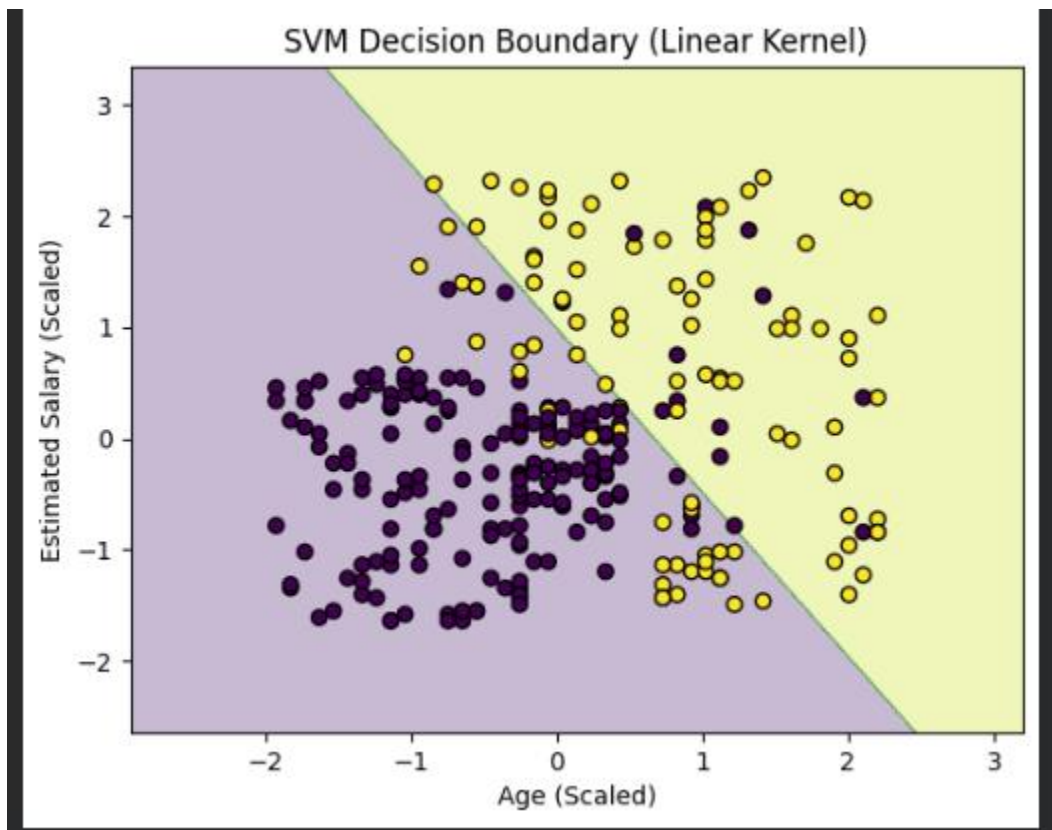
```
XX, YY = np.meshgrid(
    np.arange(x_min, x_max, 0.01),
    np.arange(y_min, y_max, 0.01)
)
```

```
Z = model.predict(np.c_[XX.ravel(), YY.ravel()])
Z = Z.reshape(XX.shape)
```

```
plt.contourf(XX, YY, Z, alpha=0.3)
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, edgecolors='k')
plt.title("SVM Decision Boundary (Linear Kernel)")
plt.xlabel("Age (Scaled)")
plt.ylabel("Estimated Salary (Scaled)")
plt.show()
```

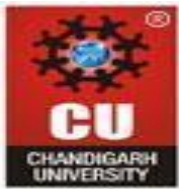
## OUTPUT:-

```
*** Accuracy: 0.85  
Confusion Matrix:  
[[72  1]  
 [17 30]]
```



### 3. Learning outcomes (What I have learnt):

- Acquire practical understanding of fundamental Python libraries including **NumPy**, **Pandas**, **Matplotlib**, and **Scikit-learn** for data preprocessing, analysis, and visualization.
- Load, preprocess, and explore a real-world classification dataset using **Pandas**, and identify relevant independent variables (Age and Estimated Salary) for predictive modeling.
- Develop conceptual understanding of **Support Vector Machine (SVM)** as a supervised learning algorithm and its role in solving binary classification problems by constructing an optimal separating hyperplane.
- Implement an SVM classifier using the **Linear Kernel**, including proper feature scaling through **Standardization** to enhance model performance and ensure fair feature contribution.
- Visualize the SVM decision boundary to examine how the model differentiates between purchasing and non-purchasing customers.



UNIVERSITY INSTITUTE *of*  
**COMPUTING**  
*Asia's Fastest Growing University*

**NAAC**  
**GRADE A+**  
ACCREDITED UNIVERSITY